Review 01 — Romallosa 2003 replication

`mcdo_dev` Phase 1   (`output/01_romallosa/`, `docs/theory/01_richards_wolf_pulse.md`)

2026-06-12

Engine and validated conventions

Richards–Wolf integrals I_0, I_1, I_2 (paper Eqs. 5–7); intensity cut $|I_0 - I_2|^2$ (x-polarized beam, scan $\perp$ polarization, $\phi = \pi/2$) — the only cut with true zeros, required by the paper's "cw contains zero points". Pulsed intensity = Eq. 10 spectral sum, **201 frequencies over the full positive spectrum** $\omega \in (0, 2\omega_c)$. This window is the critical convention: the literal FWHM-band reading of the paper's text discards 24% of the spectral weight and gives $S_{tr} = 1.020$ vs the paper's 1.062. Implemented as half-width $\min(4\sigma, \omega_c)$ in `config.omega_grid()`.

Further conventions established numerically: the paper's Fig. 2 *panel labels are swapped* (its "r $\pm 6\,\mu$m" panel is the axial profile, zeros at $\pm 3.05\,\mu$m); Fig. 3's energy fraction is a *1-D line integral* over the 201-point array (cylindrical gives 0.46 — wrong); Linfoot arrays sample in *optical coordinates* ($v \in \pm 6.70$, $u \in \pm 24.74$), which is what makes F, S NA-flat; Fig. 1 axes are $r \pm 2.0$, $z \pm 6.0\,\mu$m (scan lost the decimals).

Scorecard

Figure	paper anchor	ours	status
1	cw island contours / pulsed smooth barrel	same topology	✓
2	zeros 0.555/3.05 μm; pedestal $\approx$ 0.1 / 0.05	0.555/3.05; 0.13 / 0.049	✓
3	plateau $\approx$ 0.78; 0.65 @ 1 fs; linear $\tau \leq 2$ fs	0.777; 0.62	✓ -3% @1 fs
4	S_{ax}=1.057, S_{tr}=1.062, F≈0.96, NA-flat	1.058, 1.053, 0.96, flat	✓ $S_{tr} - 0.9\%$
5	$S(1\,\text{fs})$ on $0.97\tau^{0.04}$	0.970	✓ exact

Residuals trace to conventions the paper never documents (exact array windows, RK4 quadrature). CW anchors: transverse FWHM 472.8 nm, axial 2419 nm.

Bugs found in old mcdo (fixed on its **`cleanup`** branch, commit **`695f9da`**): spectral power $\sqrt{2}$ too narrow (double-squaring — 1 fs behaved as 1.4 fs); library window truncated at the FWHM; circular polarization cannot reproduce the zero structure.

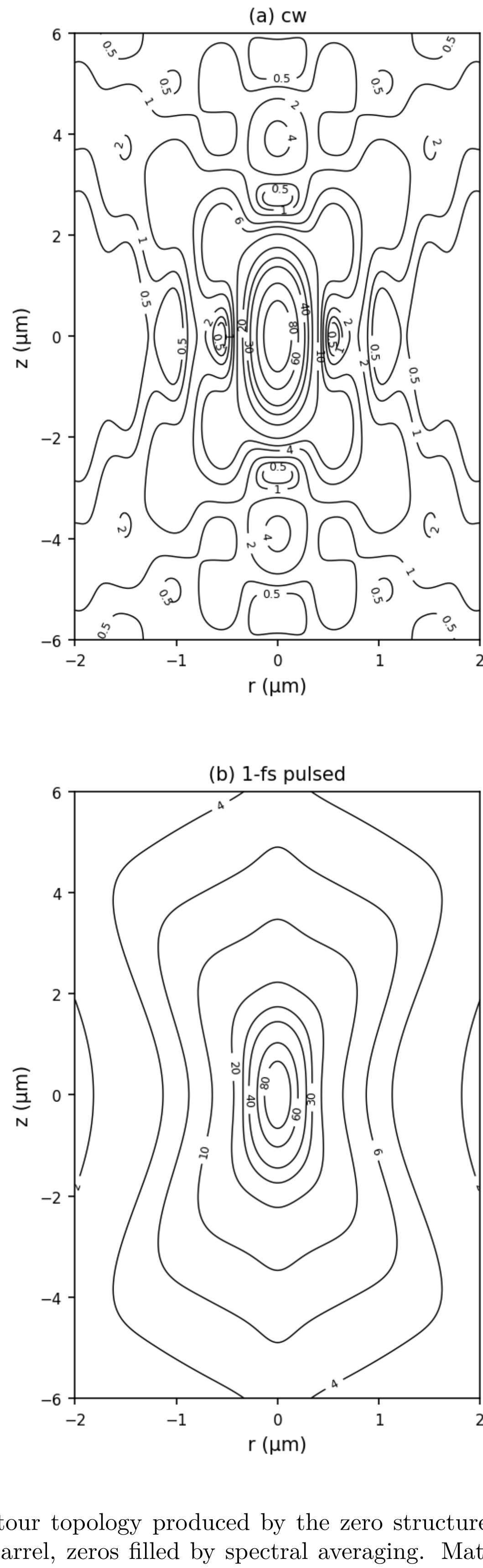

Fig. 1 — cw panel: island-contour topology produced by the zero structure (axial sidelobe islands, diagonal ridges); pulsed panel: smooth barrel, zeros filled by spectral averaging. Matches the paper's Fig. 1 layout on $r \pm 2.0$, $z \pm 6.0\,\mu$m.

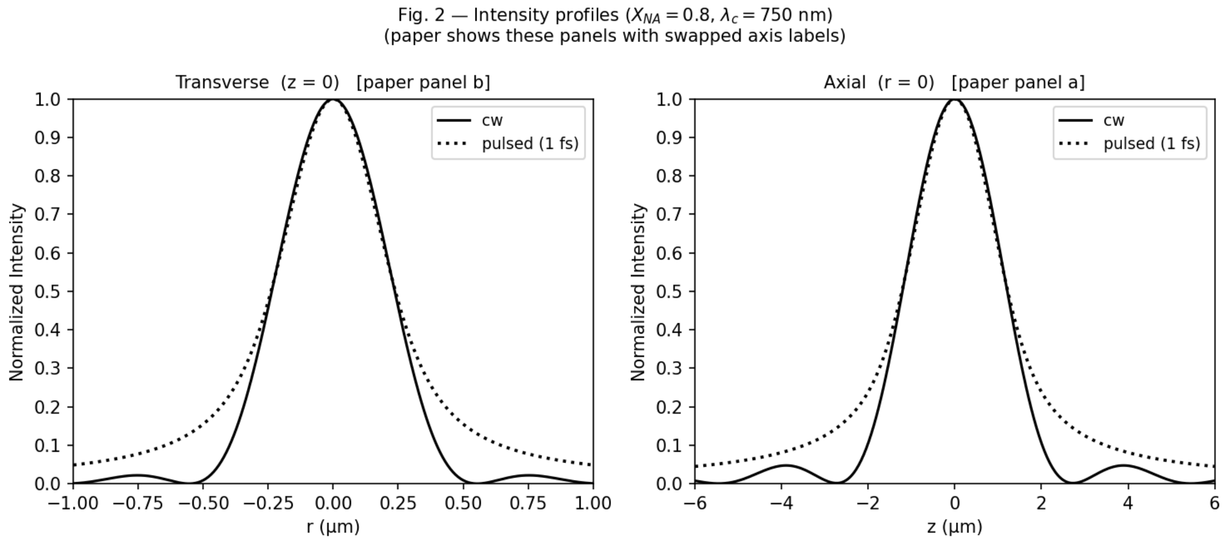

Fig. 2 — cw (solid) with true zeros at $0.555\,\mu$m (transverse) and $3.05\,\mu$m (axial); pulsed (dotted) pedestal ≈ 0.13 at the zero ring, 0.049 at $r = 1\,\mu$m (paper $\approx 0.1/0.05$). FWHM 472.8 nm cw / 479.0 nm pulsed.

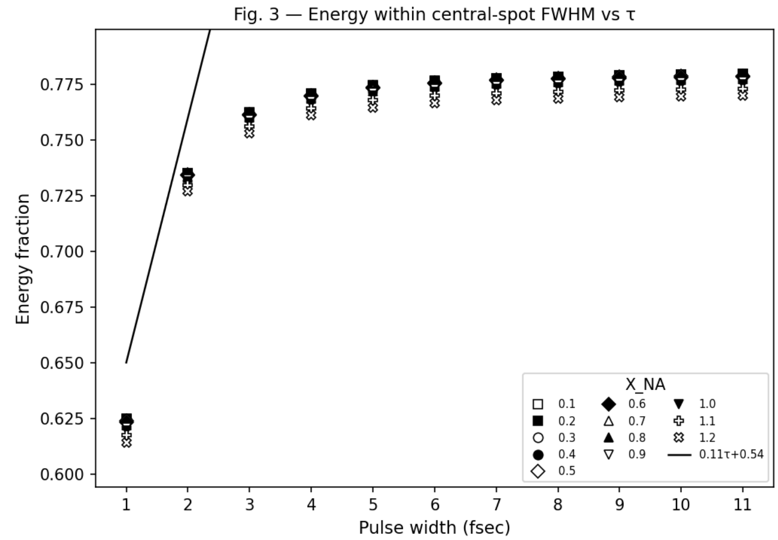

Fig. 3 — energy within the central-spot FWHM (1-D definition): plateau 0.777, sharp linear drop below $\tau = 2$ fs to 0.62 at 1 fs, reference line $0.11\tau + 0.54$ valid only $\tau \lesssim 2$ fs; tight NA clustering with $X_{NA} = 1.2$ lowest — all as published.

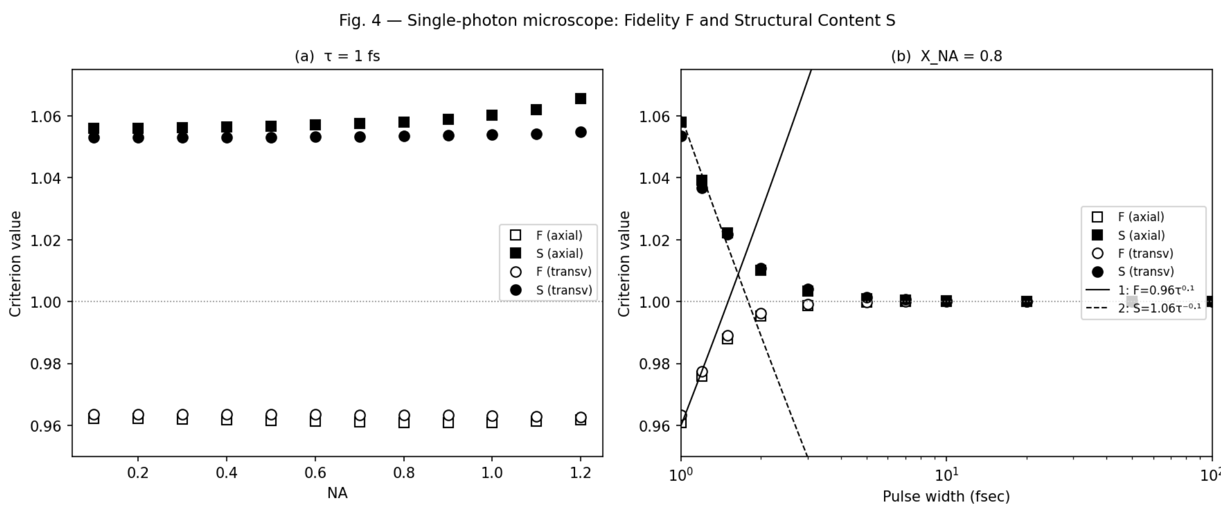

Fig. 4 — (a) F and S flat across $X_{NA} = 0.1$–1.2 at $\tau = 1$ fs (S_{ax} mean 1.058 vs paper 1.057 ± 0.0018); (b) sudden deterioration below $\tau = 1.2$ fs, data hugging the caption's $F = 0.96\tau^{0.1}$, $S = 1.06\tau^{-0.1}$ near 1–2 fs.

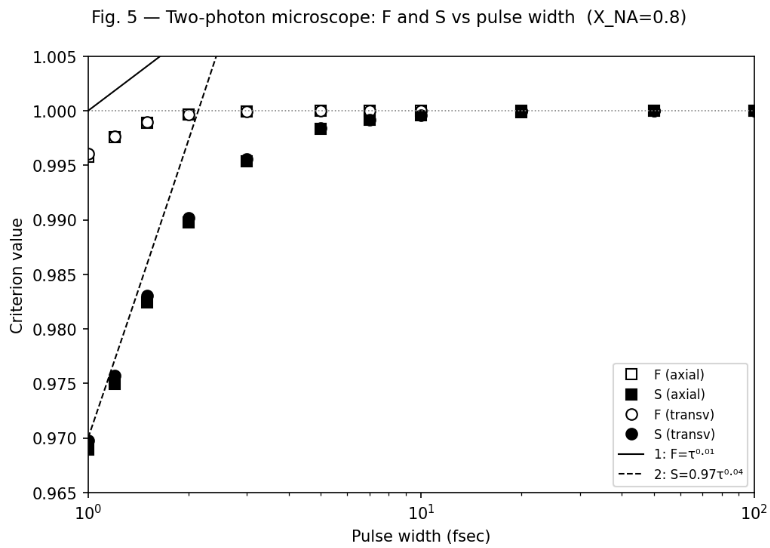

Fig. 5 — two-photon PSF = (normalized $|E|^2)^2$: $S(1\,\mathrm{fs}) = 0.970$, exactly on the caption's $0.97\tau^{0.04}$; two-photon deterioration ($\approx 3\%$) weaker than single-photon (≈ 5–6%) — the paper's headline conclusion, reproduced.